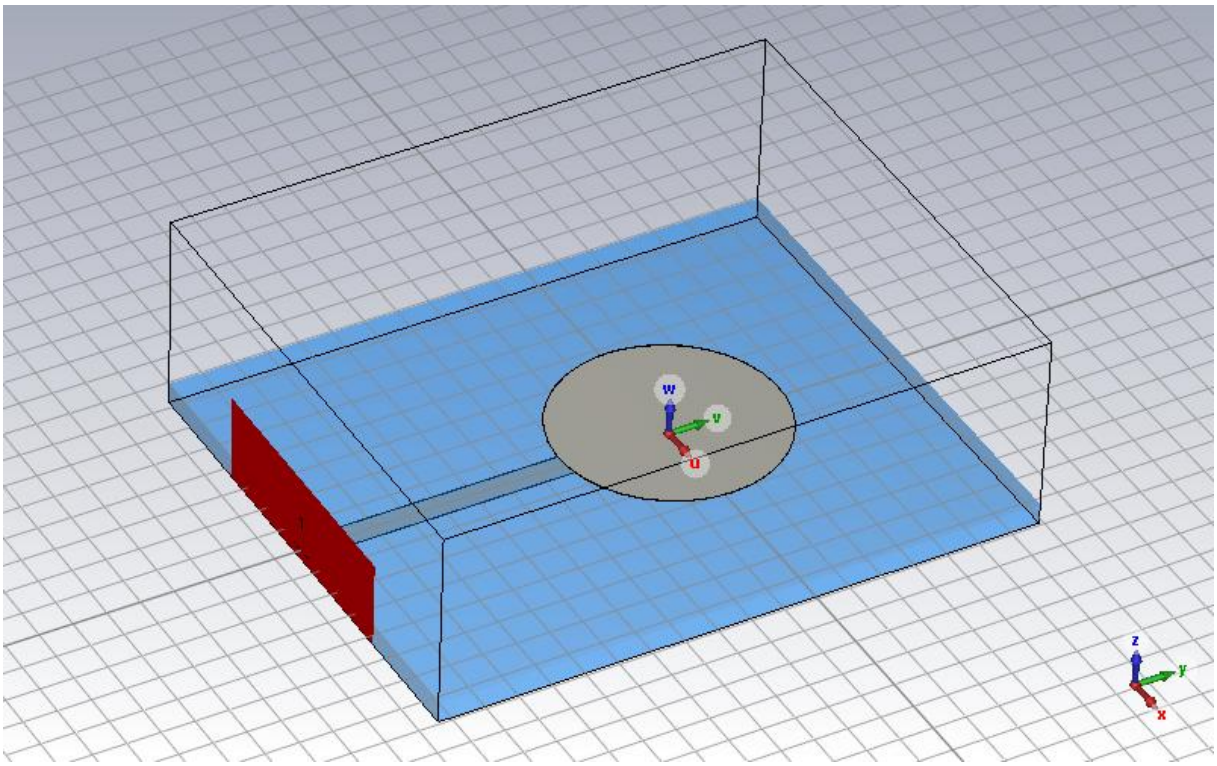
Circular Patch Antenna using CST Studio Suite

The antenna shows a broadside radiation pattern with a maximum directivity of approximately 7.09 dBi at 2.92 GHz. Maximum gain approximately 6.76 dBi for the third mesh. The radiation is predominantly directed along the z-axis, with minimal back radiation. The high radiation efficiency -0.3289dB(93%) indicates low material losses, while the total efficiency result -0.7368 dB (84%) shows minor mismatch losses.

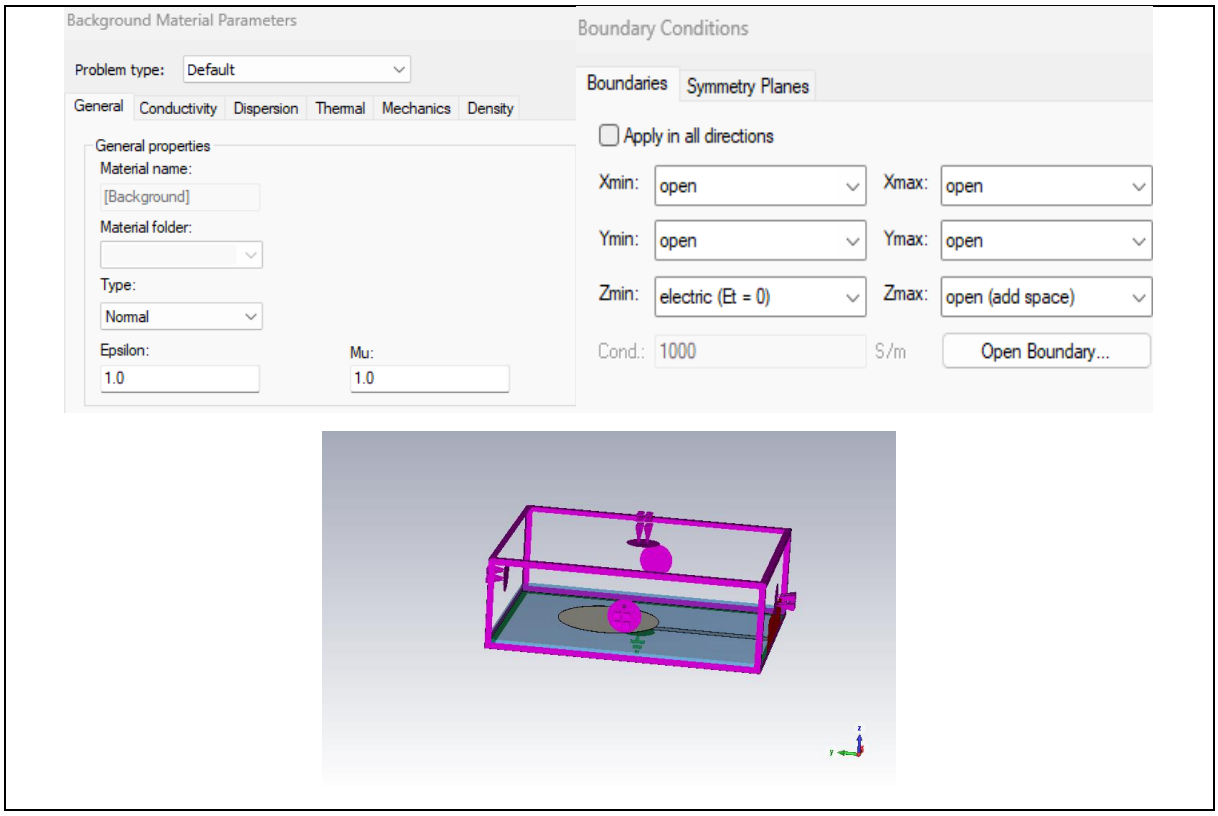
Design & Components



This project presents the design and simulation of a circular stripline TL patch antenna. The antenna consists of a circular Perfect Electric Conductor (PEC) patch positioned on a dielectric substrate with a relative permittivity of $\varepsilon_r = 2.62$. The patch is fed to a 50 Ω stripline, which is located between the patch and the ground plane.

Boundary Conditions

The ground plane is modelled using an electric boundary condition ($E_t = 0$), ensuring perfect reflection of tangential electric fields. The surrounding environment is defined as free space ($\varepsilon_r = 1$). To accurately model radiation, open boundary conditions are applied along the X_{min}, X_{max}, Y_{min}, and Y_{max} directions. The Z_{max} boundary is set to Open (add space) to allow sufficient space for radiation in the dominant direction.



The antenna is excited with a waveguide port defined at the of the strip feedline. To skip the S-parameter phase shift introduced by the feed line, the port reference plane is de-embedded inside the structure.

Meshing

Mesh adaptation is enabled based on: S-parameters and Maximum gain

Two field monitors are defined at the resonant frequency to visualize: Surface current distribution and Far-field radiation pattern

Parameter	Value
Patch radius	17 mm
Patch thickness	0.05 mm
Stripline width	5 mm
Stripline length	56 mm
Stripline thickness	0.05 mm

Core antenna performance

S11 Return Loss (3 mesh passes)

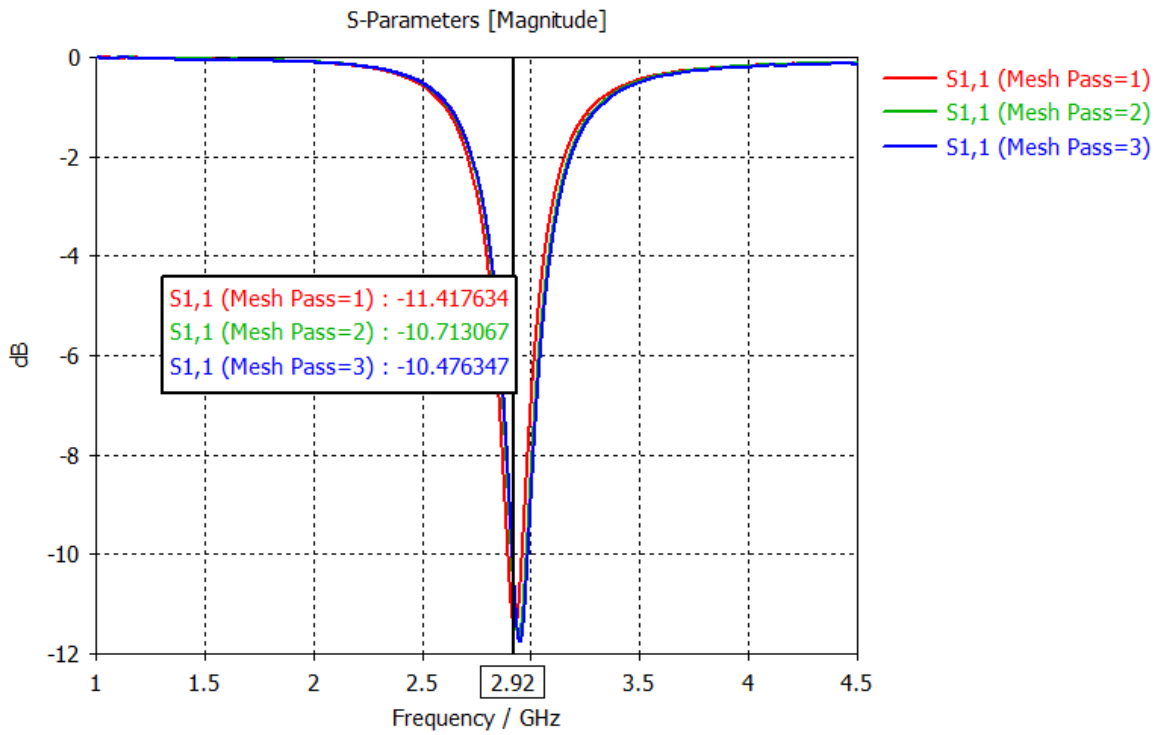
S11 is the return loss, how much of the transmitted signal is reflected back from the antenna. All three mesh passes show a clear resonance dip at 2.92GHz. The three curves are nearly identical confirming mesh convergence. The convergence from pass 1 to pass 3 shows the solution is stabilising, the small shift from −11.42 to −10.48 dB is only 0.94 dB, well within acceptable simulation tolerance.

Bandwidth calculation:

The −10 dB bandwidth (where S11 < −10 dB) is the standard measure. Reading from the plot the curve crosses −10 dB at approximately **2.75 GHz** (lower edge) and **3.10 GHz** (upper edge):

$$BW_{-10dB}=3.10-2.75=350 \text{ MHz}$$

$$\text{Fractional BW} = \frac{350}{2920}\ x\ 100 = \ 12\%$$

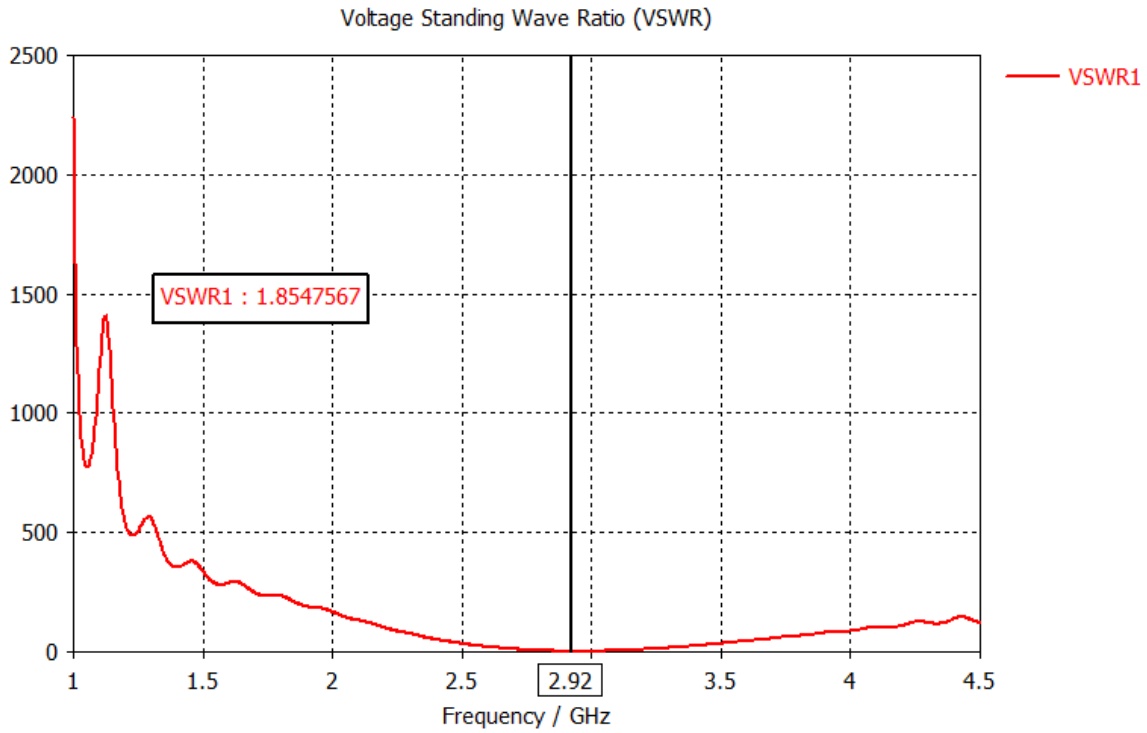


S11 return loss magnitude across three adaptive mesh passes at the resonant frequency of 2.92 GHz. The antenna achieves a minimum S11 of −10.48 dB (Mesh Pass 3) at resonance, satisfying the standard −10 dB matching criterion. The −10 dB impedance bandwidth spans approximately

2.75–3.10 GHz, yielding a bandwidth of 350 MHz (12% fractional bandwidth centred at 2.92 GHz). The close agreement between all three mesh passes confirms simulation convergence and validates the accuracy of the final result.

VSWR

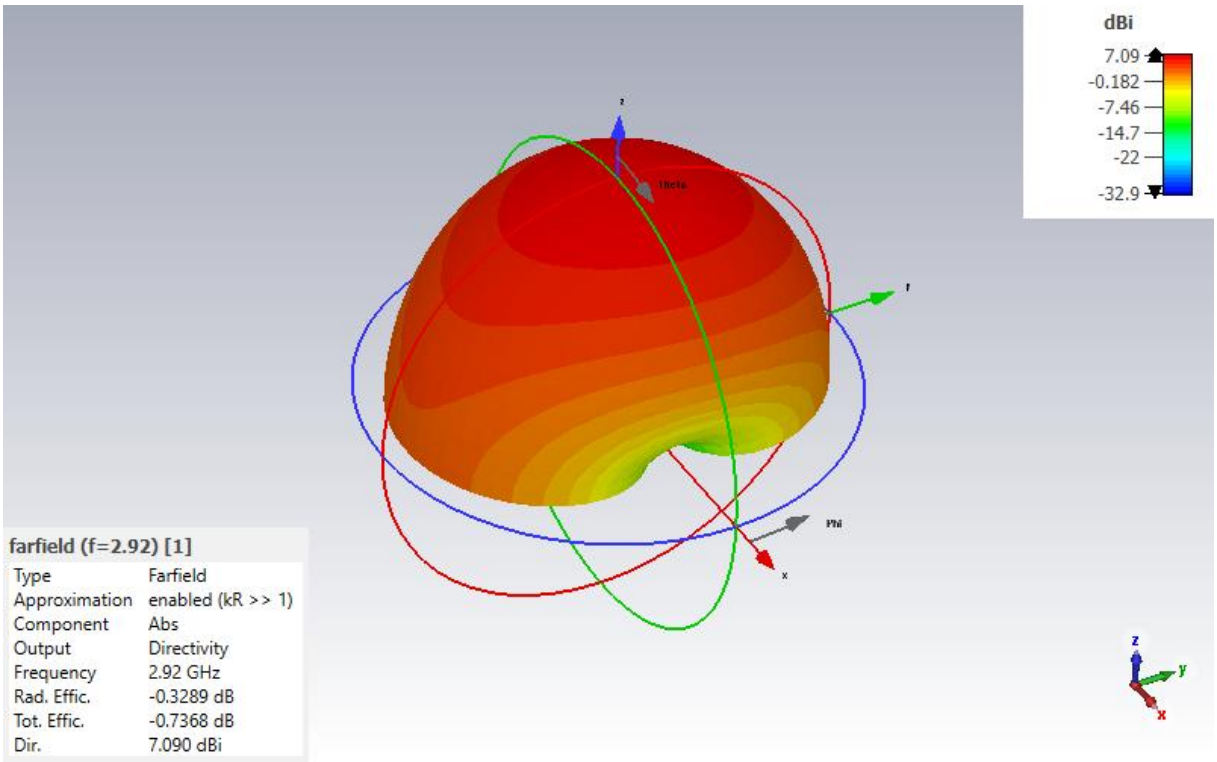
Voltage Standing Wave Ratio (VSWR) of the circular patch antenna across 1–4.5 GHz. The VSWR exhibits a sharp minimum of 1.855 at the resonant frequency of 2.92 GHz, consistent with the S11 result of −10.48 dB. The VSWR across the non-resonant frequency range (1.0–2.5 GHz) confirms the narrowband nature of the patch antenna, operating as expected outside its design frequency. The VSWR < 2 condition, commonly used as the antenna matching criterion, is satisfied in the vicinity of 2.92 GHz.



A VSWR of 1.855 corresponds to a reflection coefficient of $|\Gamma| = 0.30$, meaning 30% of the incident power is reflected and 70% is accepted by the antenna.

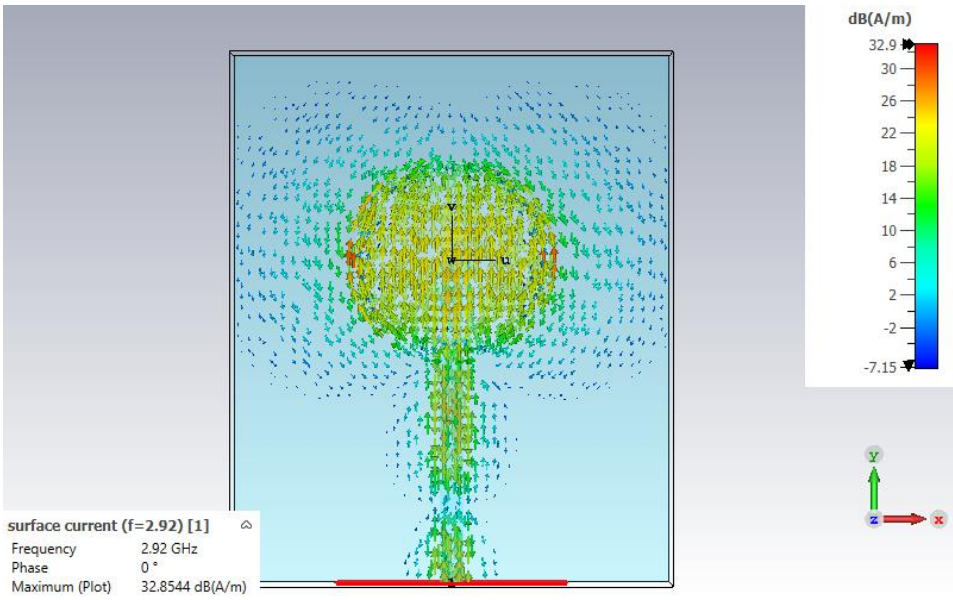
Farfields\farfield (f=2.92) [1]

Three-dimensional far-field radiation pattern at the resonant frequency of 2.92 GHz. The pattern exhibits a broadside radiation characteristic with the main lobe directed along the +z axis, perpendicular to the patch plane. Maximum directivity is 7.09 dBi. The pattern is nearly omnidirectional in the azimuthal plane (xy-plane) with minimal back radiation, consistent with theoretical predictions for a circular microstrip patch antenna.



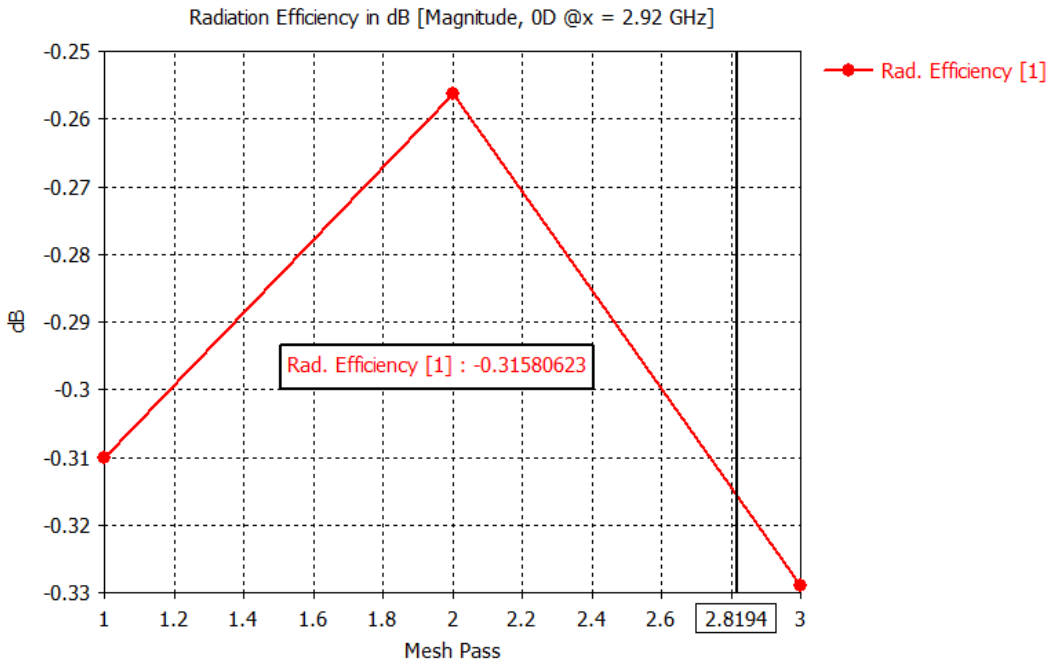
Surface Current Distribution (f=2.92) [1]

Surface current distribution on the circular patch at 2.92 GHz. The current distribution reveals the dominant TM_{11} resonant mode, with maximum current density concentrated along the patch edge in the direction of the feed. The symmetrical current pattern confirms that the antenna is operating in the intended fundamental mode, which is responsible for the broadside radiation pattern observed below



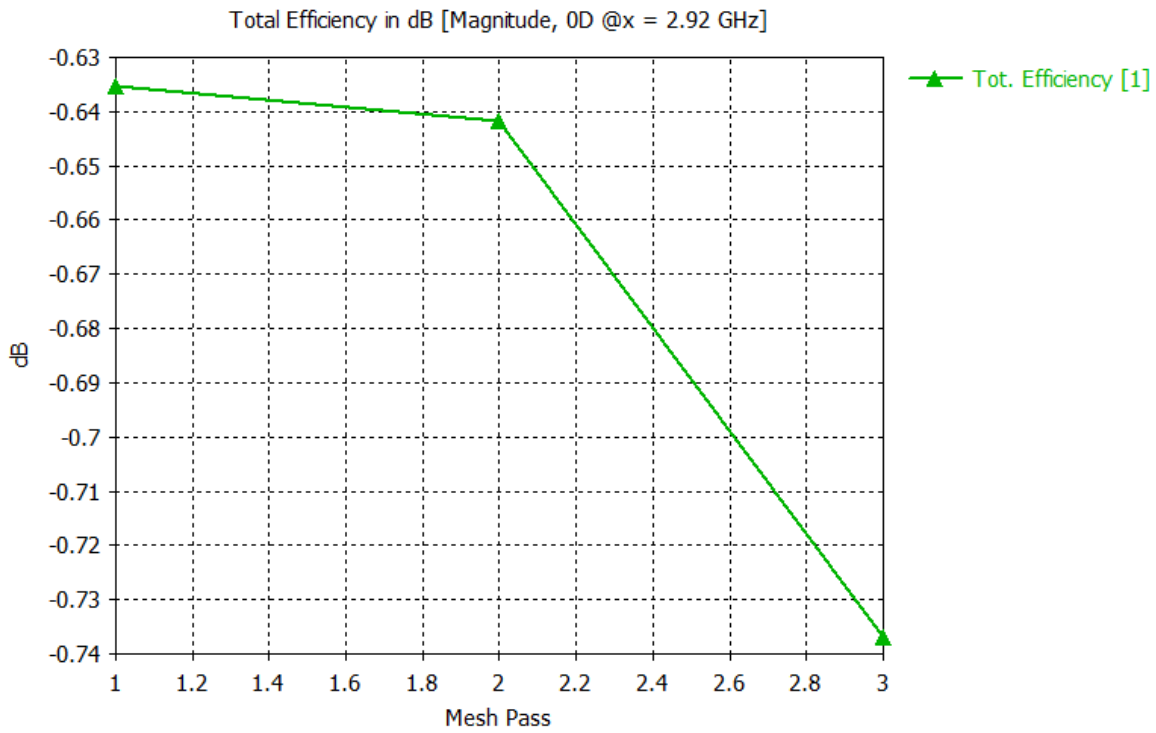
1D Results\Efficiencies\Rad. Efficiency [1]

Radiation efficiency as a function of mesh passes. At the resonant frequency of 2.92 GHz, the radiation efficiency reaches −0.33 dB (93%), indicating that 93% of the power accepted by the antenna is radiated into free space, with only 7% dissipated as ohmic and dielectric losses. This high efficiency is consistent with the use of low-loss PEC conductors and a low-loss substrate.



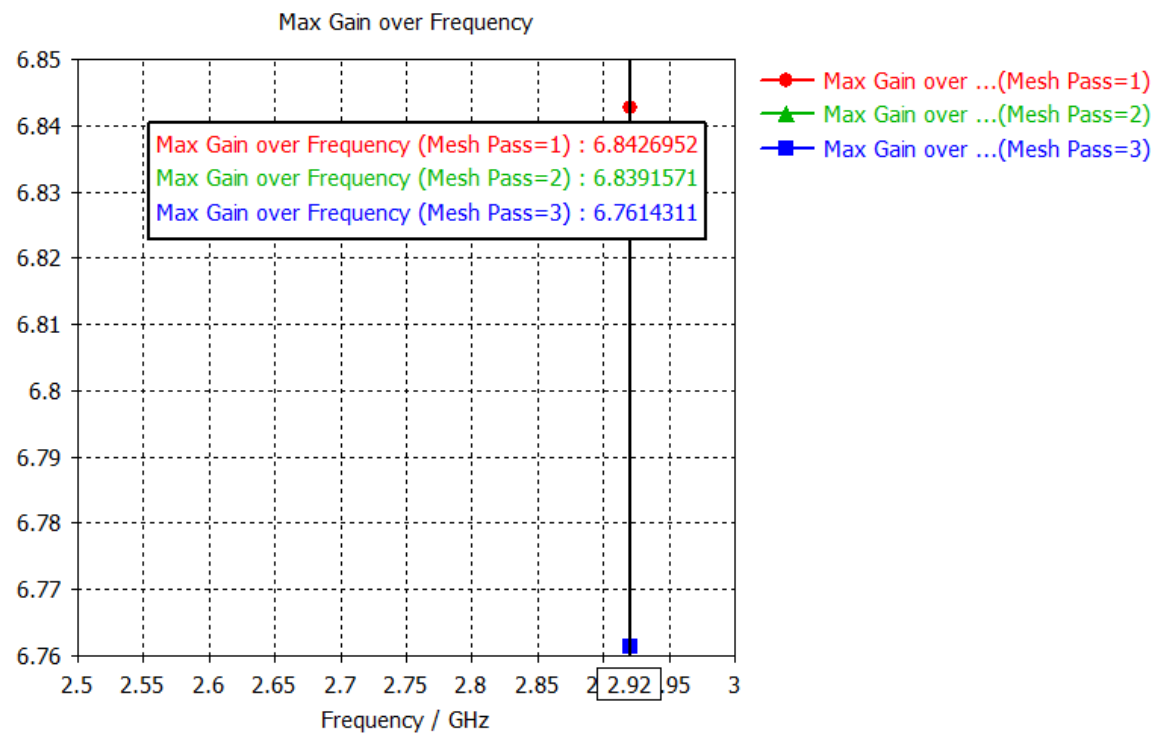
1D Results\Efficiencies\Tot. Efficiency [1]

Total efficiency as a function of mesh passes. At 2.92 GHz, the total efficiency is −0.74 dB (84%), accounting for both radiation efficiency (93%) and impedance mismatch losses at the feed port. The difference between radiation and total efficiency (9%) represents the reflected power due to impedance mismatch, which can be further reduced through feed optimisation.



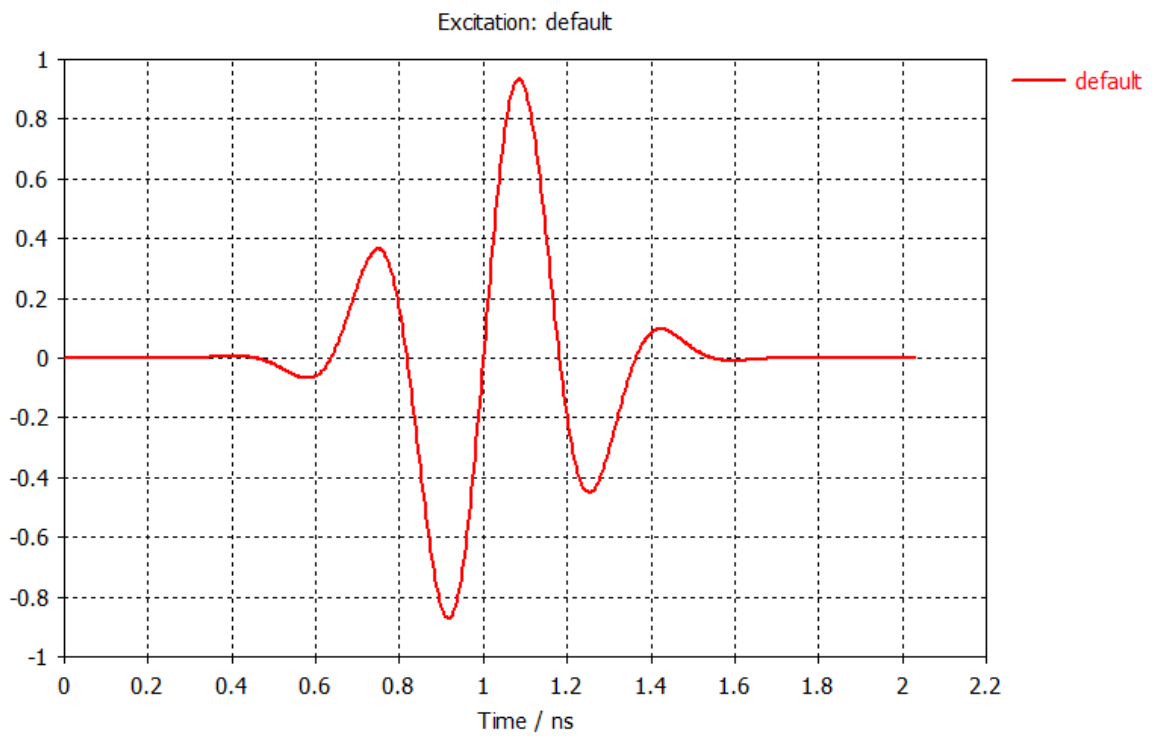
Tables\1D Results\Max Gain over Frequency

Maximum gain values versus frequency. Numerical gain data extracted at discrete frequency points enabling precise identification of the peak gain frequency and the −1 dB and −3 dB gain bandwidth. Peak gain of 6.76 dBi is recorded at 2.92 GHz.



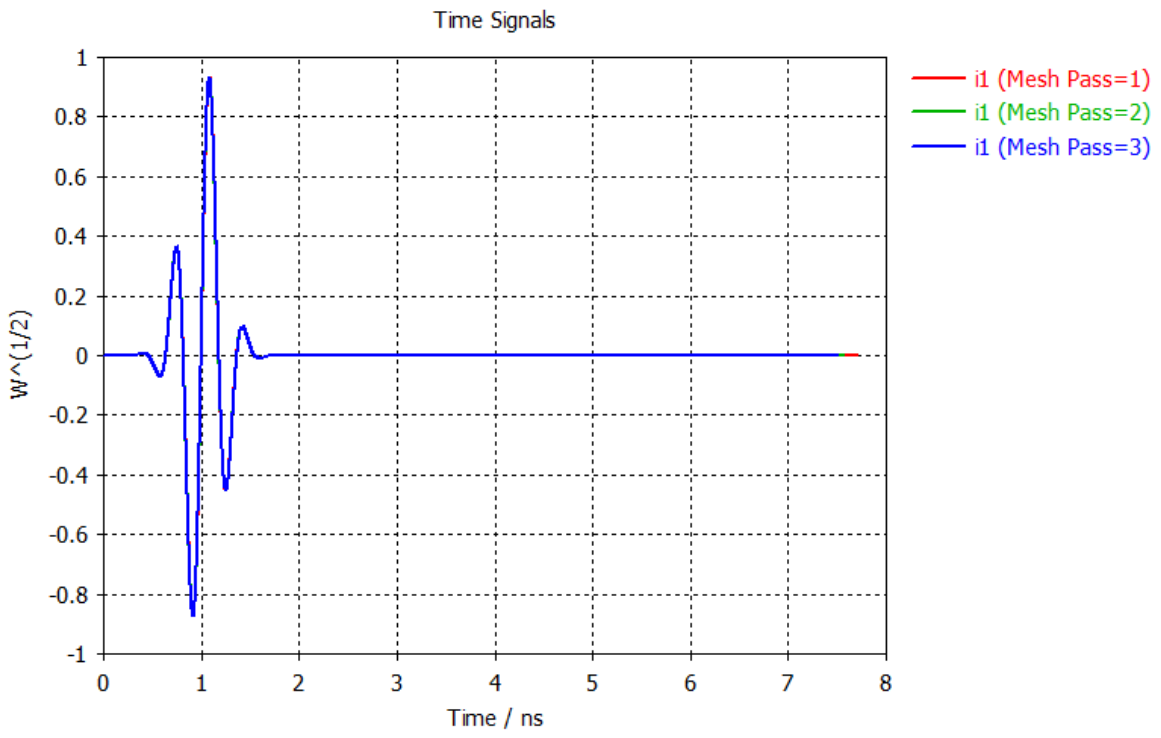
Excitation Signals

Time-domain excitation signal applied at Port 1. The Gaussian pulse covers the full simulation frequency range, providing broadband excitation necessary for frequency-domain post-processing via Fourier transform. The pulse amplitude and width are automatically determined by CST based on the defined frequency range.



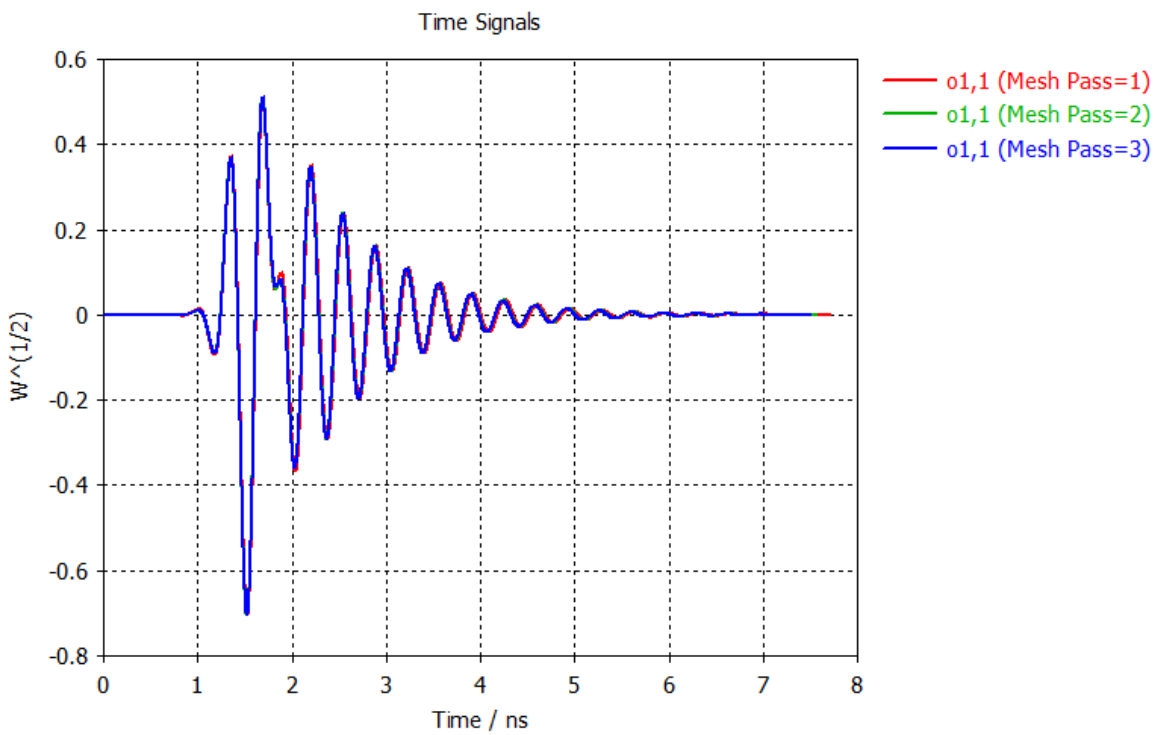
1D Results\Port signals\i1

Time-domain excitation signal i1 at Port 1 across three adaptive mesh passes. The broadband Gaussian pulse is centred at approximately t = 1 ns with a peak amplitude of 0.9 $W^{(1/2)}$. Complete signal decay to zero by t = 3 ns confirms that all transient electromagnetic fields have dissipated within the simulation window, ensuring accurate and artefact-free Fourier transformation to the frequency domain. The near-identical pulse profiles across all three mesh passes confirm consistent port excitation throughout the adaptive refinement process.



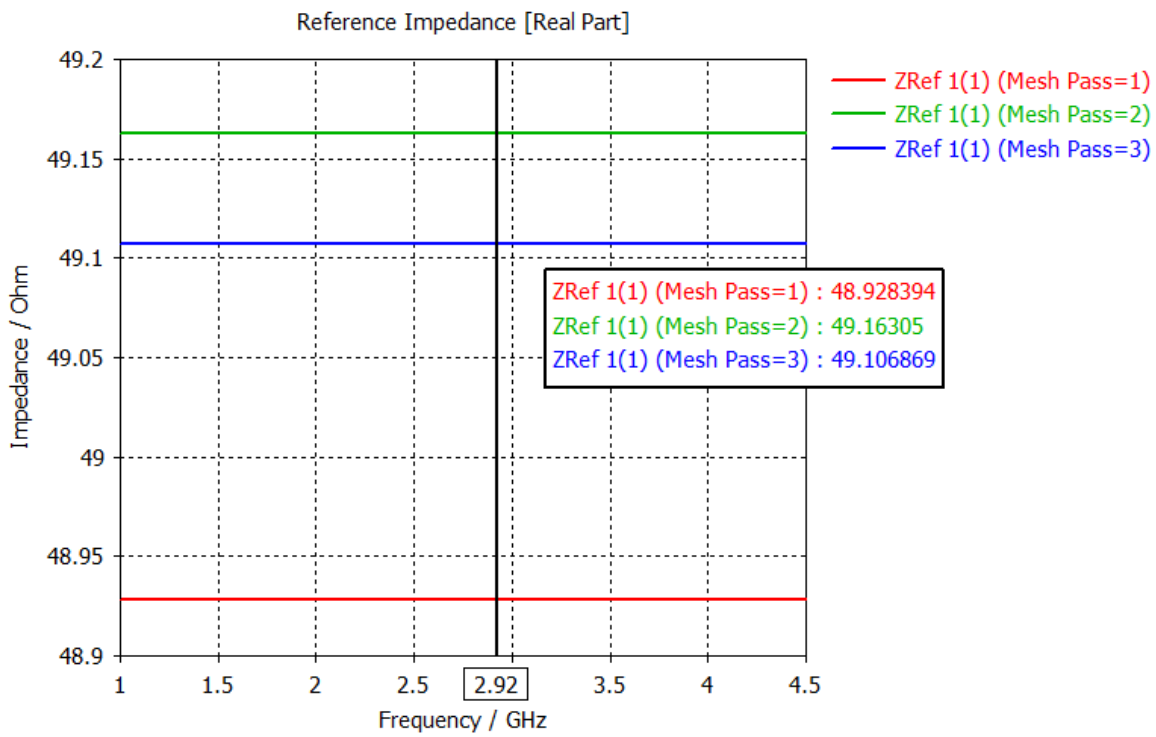
1D Results\Port signals\o1,1

Time-domain reflected signal at Port 1 (o1,1). This represents the wave reflected from the antenna input. The decay of the signal toward zero indicates that the simulation has run sufficiently long for the fields to reach steady state, confirming reliable S-parameter extraction.



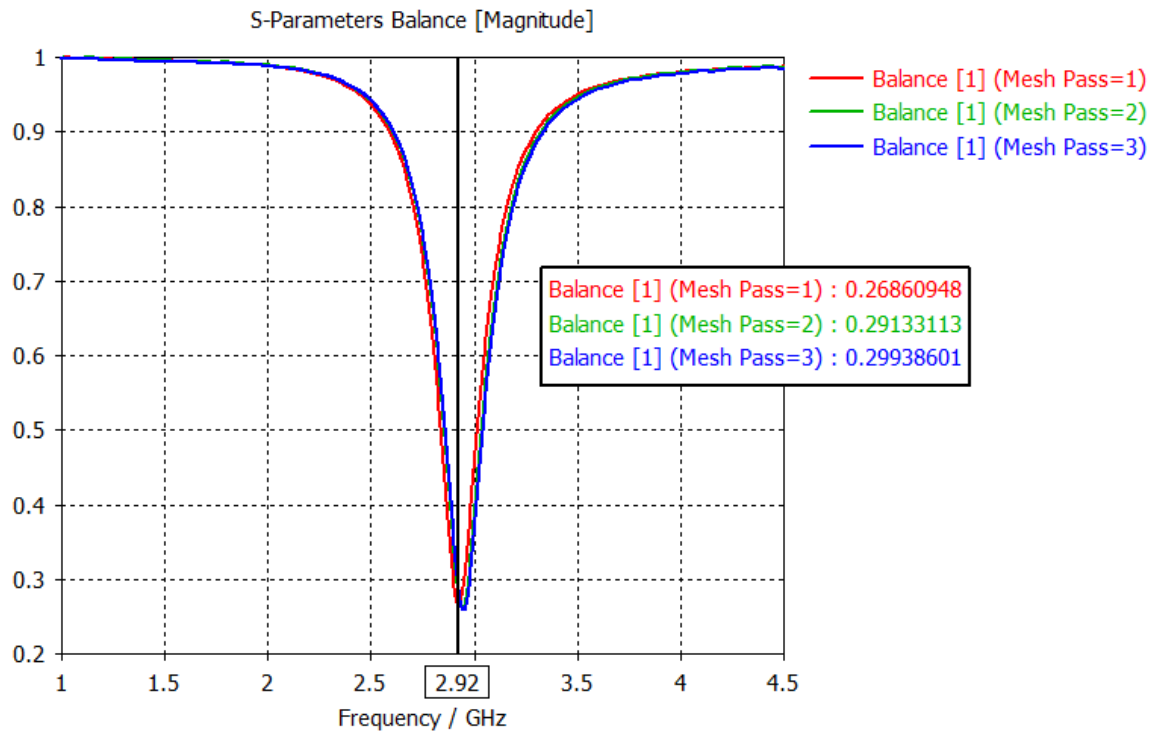
1D Results\Reference Impedance\ZRef 1(1)

Reference impedance of Port 1 as a function of frequency. The port is defined with a 50 Ω reference impedance, consistent with standard RF system design. S-parameters are normalised to this reference throughout the simulation.



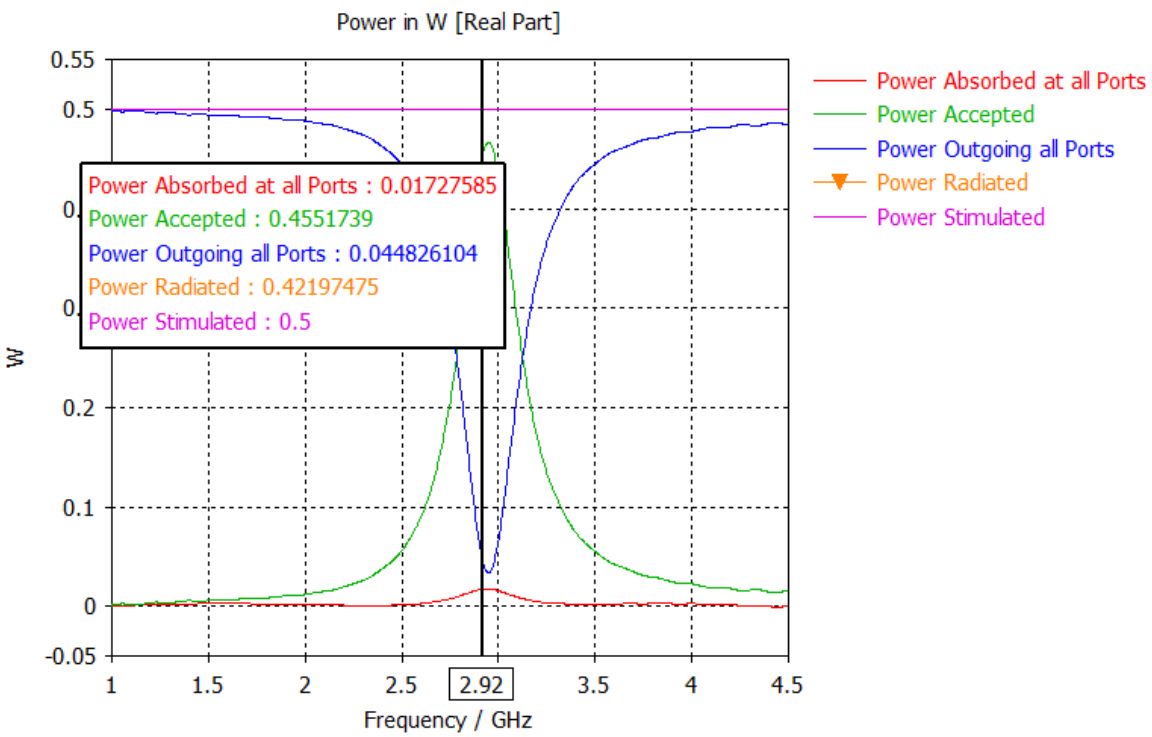
1D Results\Balance\Balance [1]

Power balance across the simulation domain. The balance confirms conservation of energy — the sum of accepted power, radiated power, and absorbed power should equal unity. A balance close to 1 (0 dB) across the frequency range validates the accuracy of the simulation mesh and boundary conditions.



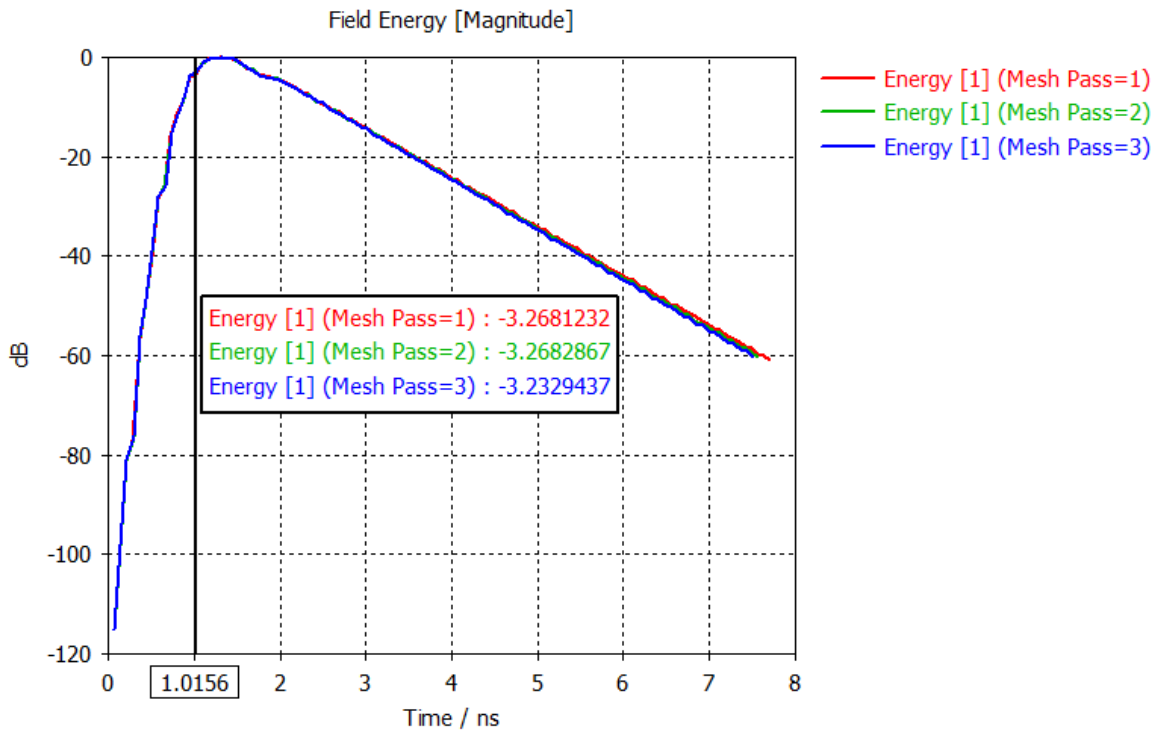
1D Results\Power\Excitation [1]

Frequency spectrum of the excitation power delivered to Port 1. The flat profile across the frequency band confirms uniform power delivery, ensuring that gain and efficiency results are not distorted by frequency-dependent excitation weighting



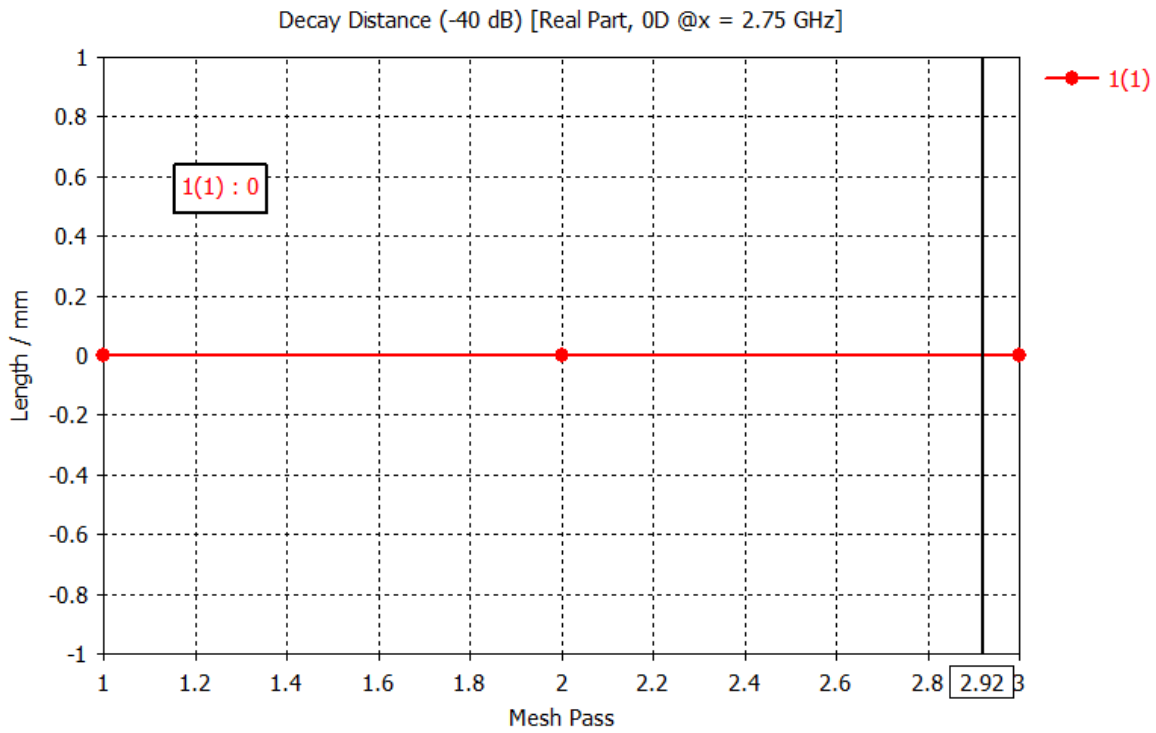
1D Results\Energy\Energy [1]

Time-domain energy decay within the simulation volume. The exponential decay of stored electromagnetic energy confirms that the simulation has reached convergence — all transient fields have been radiated or absorbed before the simulation ends, ensuring accurate frequency-domain results.



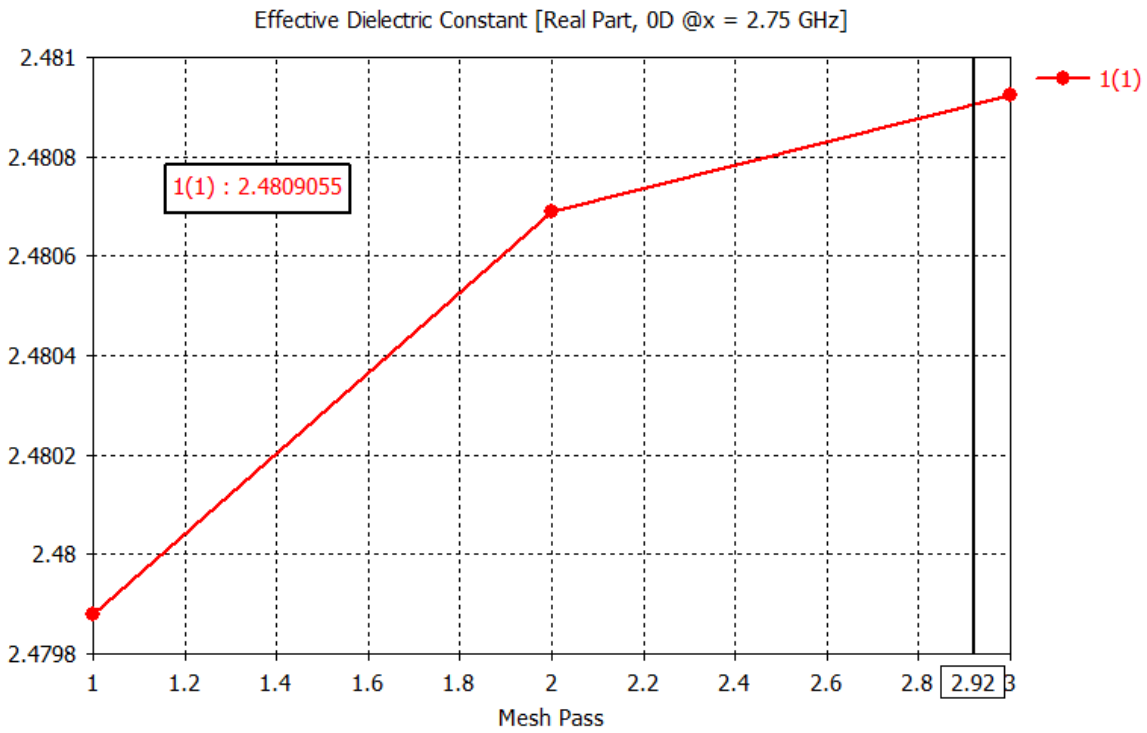
1D Results\Port Information\Distance (-40 dB)\1(1)

Port mode distance to −40 dB decay. This confirms that the waveguide port is placed sufficiently far from the antenna structure that higher-order modes have decayed to −40 dB, ensuring only the fundamental TEM mode is active at the port reference plane. This validates the accuracy of the S-parameter de-embedding.



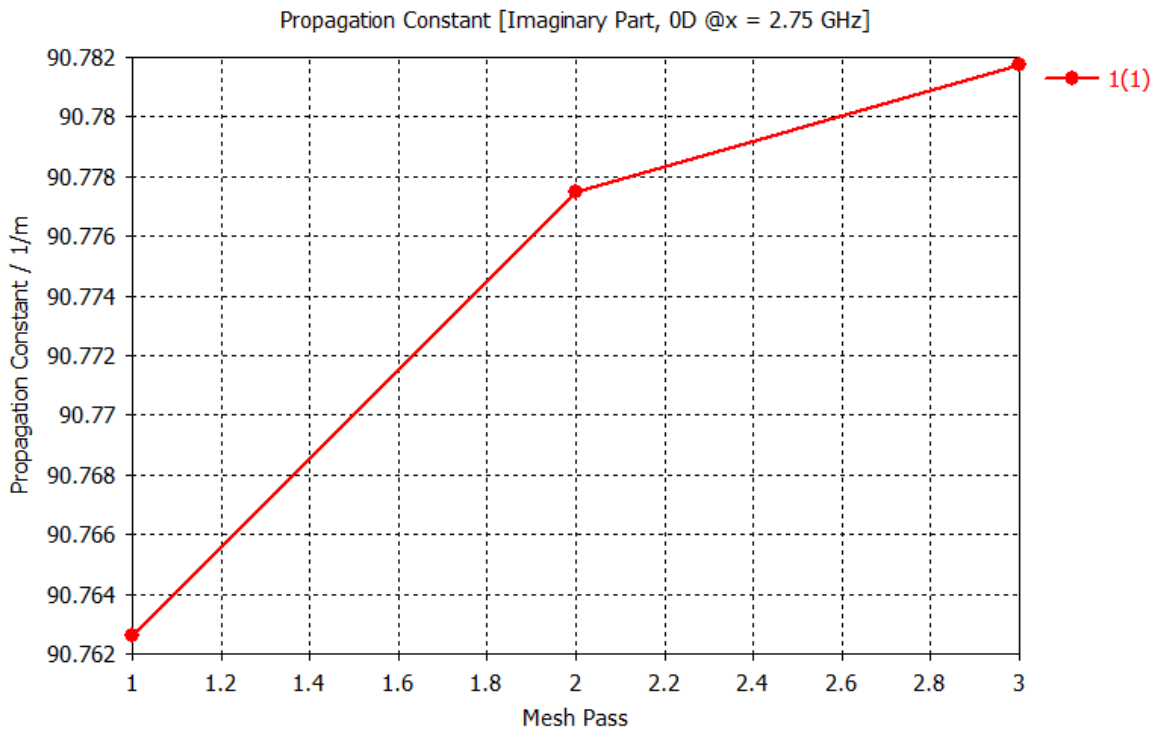
1D Results\Port Information\Effective Dielectric Constant\1(1)

Effective dielectric constant of the stripline feed as a function of frequency. The effective permittivity εeff accounts for the mixed dielectric environment of the stripline between substrate and ground plane. The relatively flat profile with frequency confirms low dispersion in the feed structure across the simulated band.



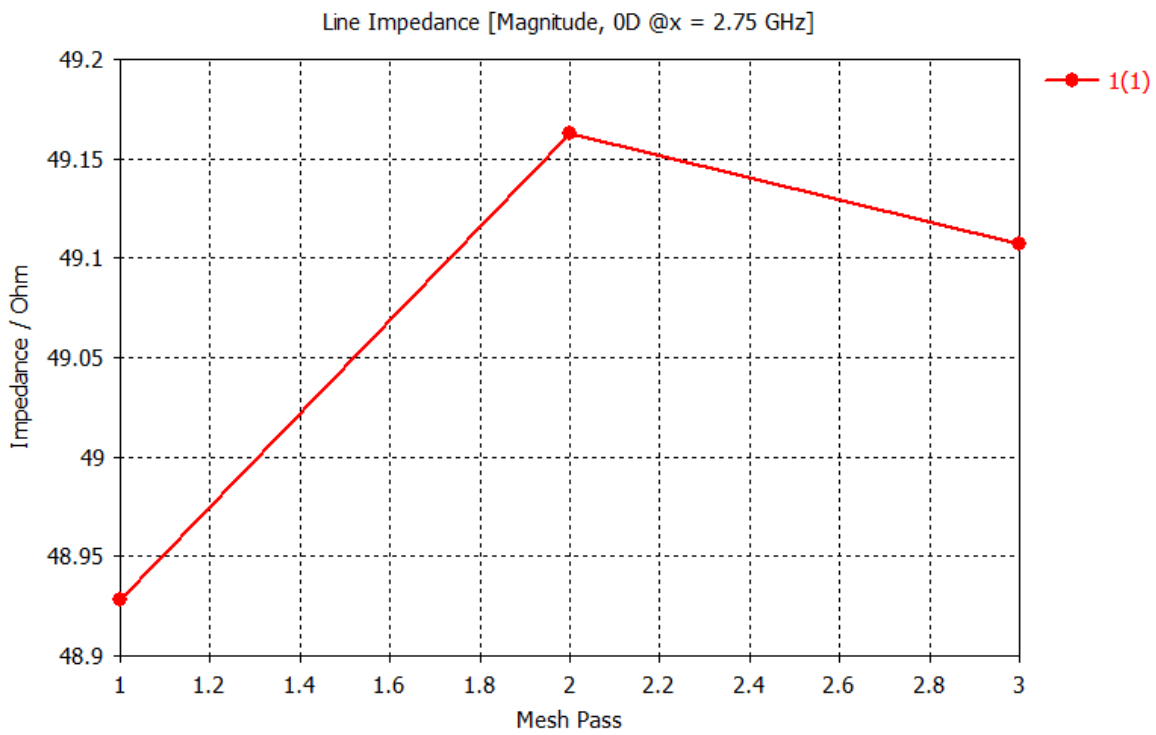
1D Results\Port Information\Gamma\1(1)

Propagation constant (Gamma) of the stripline feed port as a function of frequency. The imaginary part (phase constant β) increases linearly with frequency as expected for a low-dispersion TEM transmission line. The real part (attenuation α) remains small, confirming low feed-line losses.



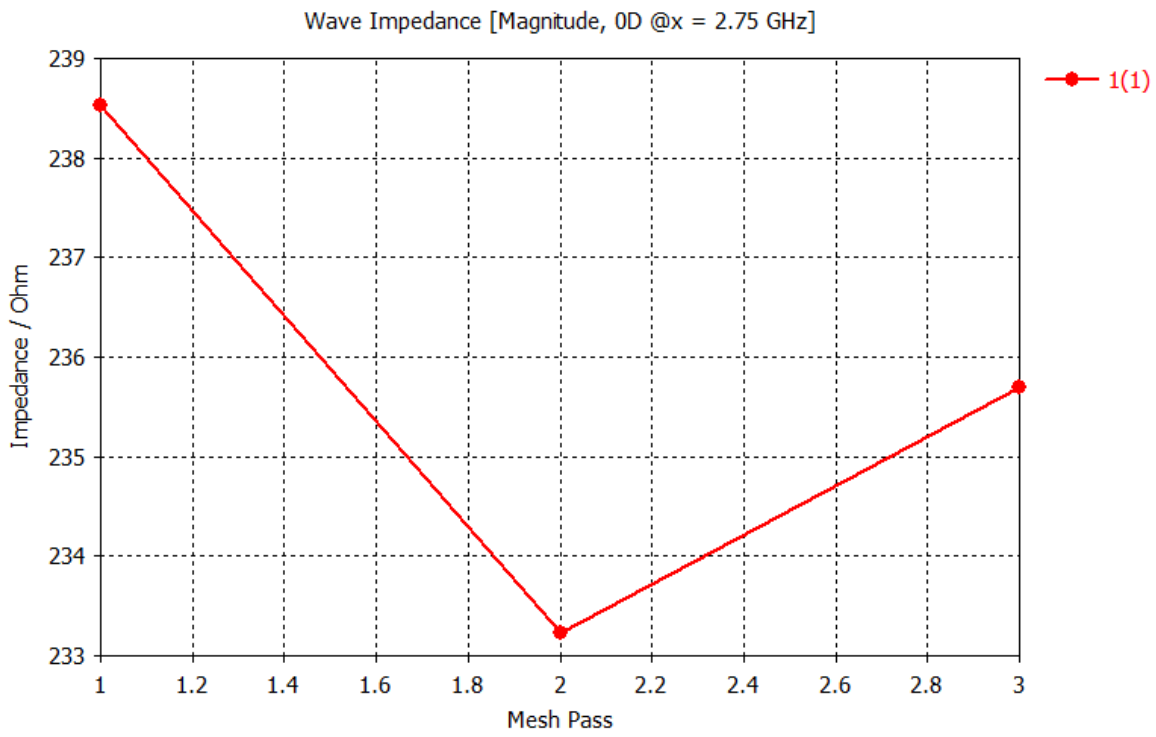
1D Results\Port Information\Line Impedance\1(1)

Characteristic line impedance of the stripline feed as a function of frequency. The impedance remains close to the target 50 Ω across the band, confirming correct feed geometry and validating the stripline width of 5 mm for the given substrate parameters (εr = 2.62).



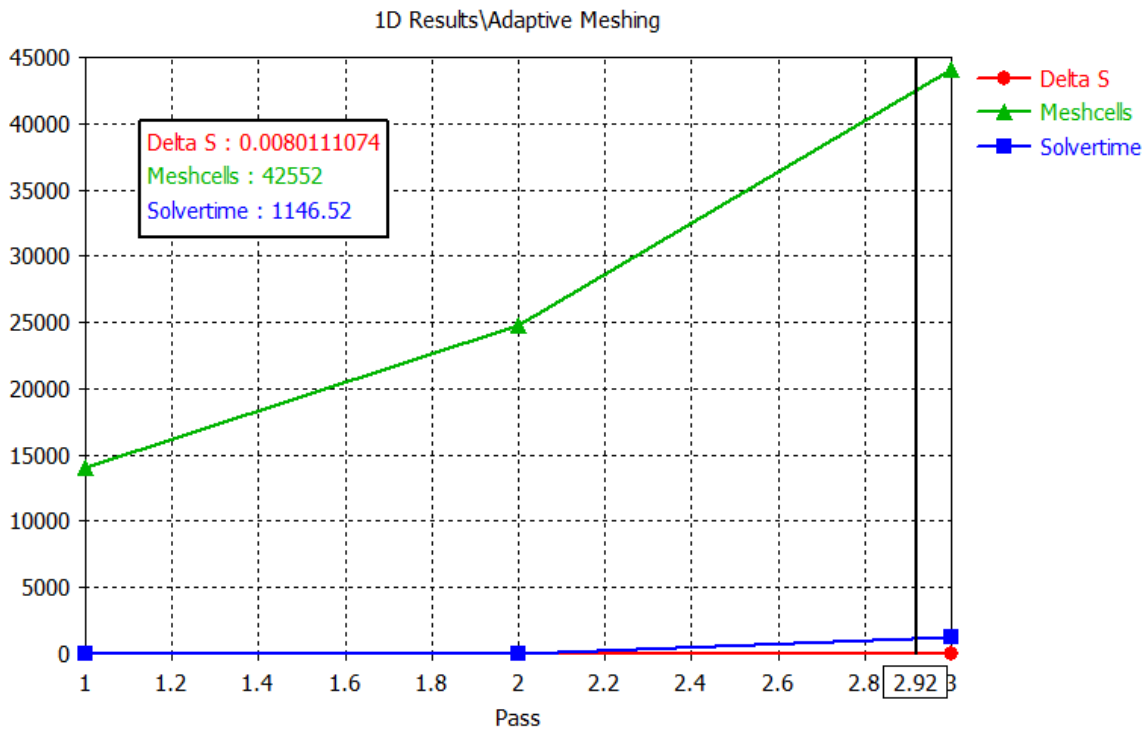
1D Results\Port Information\Wave Impedance\1(1)

Wave impedance of the dominant port mode as a function of frequency. This represents the ratio of transverse electric to magnetic field components at the port cross-section, confirming TEM-mode propagation in the stripline feed consistent with the theoretical free-space impedance scaled by the effective permittivity.



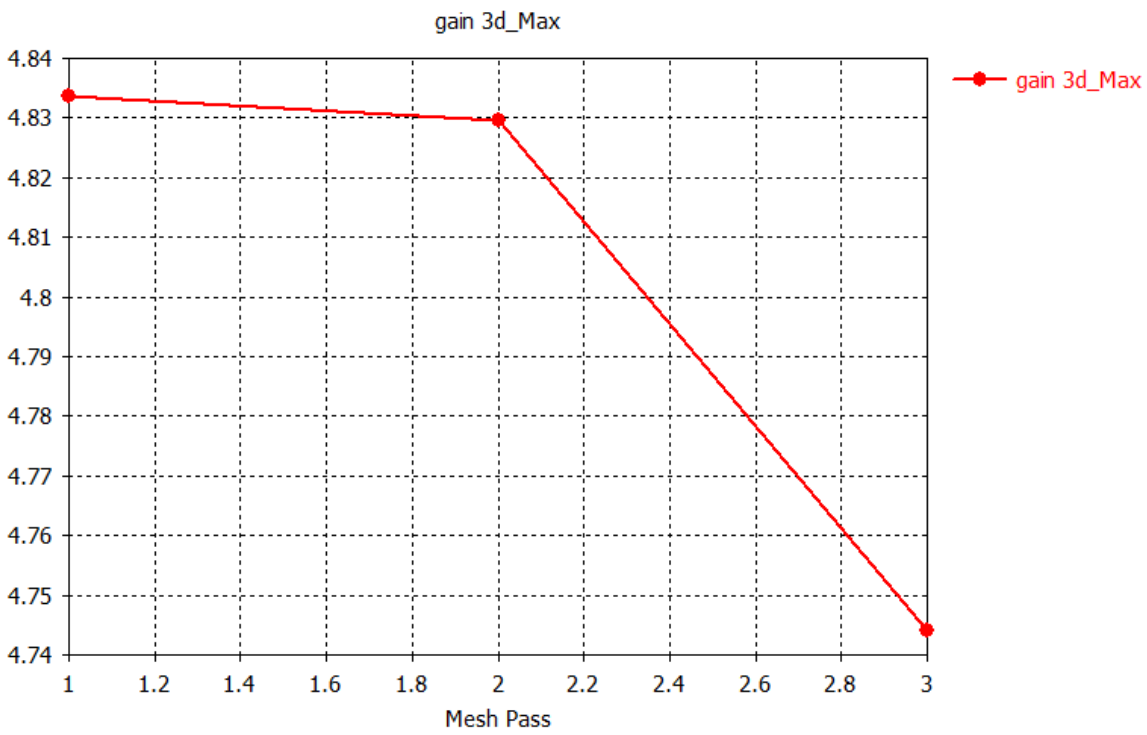
1D Results\Adaptive Meshing

Adaptive mesh refinement convergence history. Three mesh passes were performed with adaptation based on S-parameters and maximum gain. The convergence of the S11 and gain values between successive passes confirms that the final mesh is sufficiently dense to accurately resolve the surface currents on the circular patch and the near-field distribution. Final mesh produces maximum gain of 6.76 dBi.



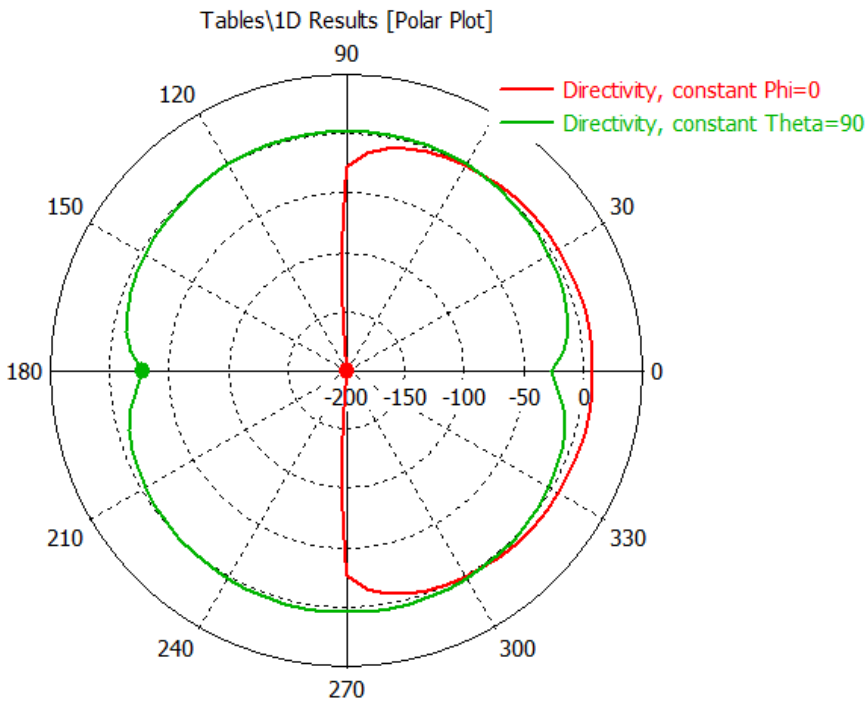
Tables\0D Results\gain 3d_Max

Maximum 3D gain as a function of mesh adaptation pass. The gain converges to 6.76 dBi by the third mesh pass, confirming simulation accuracy. The gain is 0.33 dB lower than the directivity (7.09 dBi), consistent with the 93% radiation efficiency, the small difference representing conductor and dielectric losses in the structure.



Tables\1D Results (E-plane and H-plane Directivity)

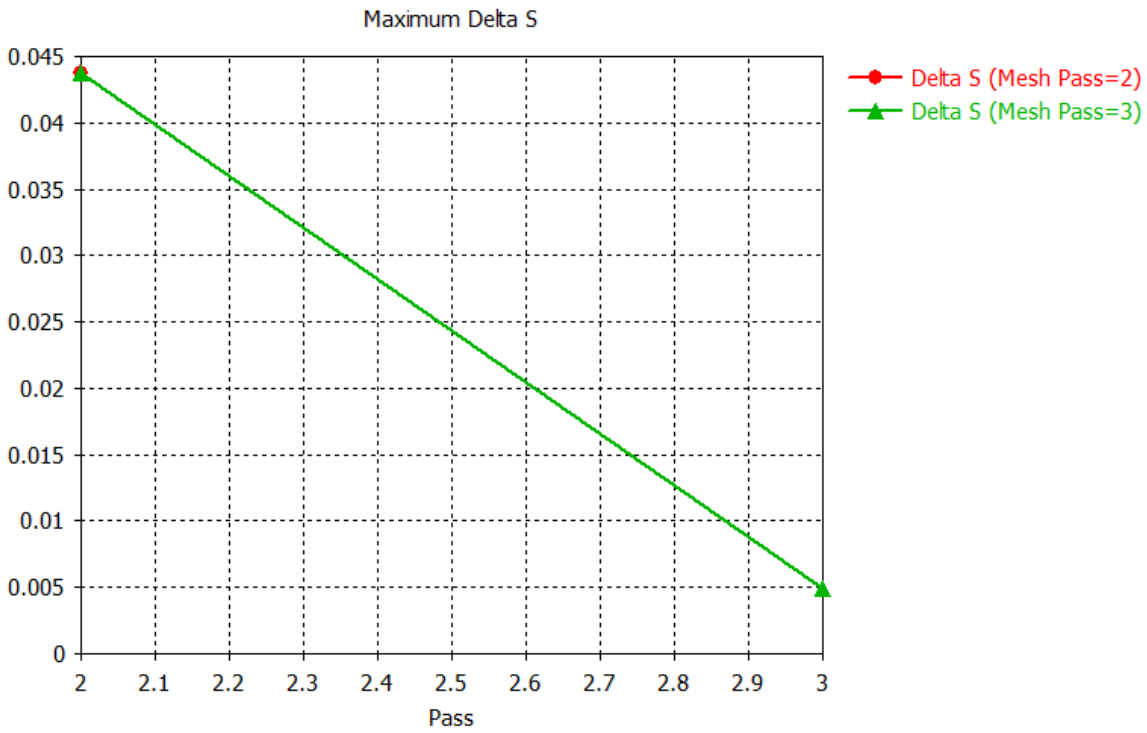
Two-dimensional polar plot of directivity at the resonant frequency of 2.92 GHz showing the E-plane cut (Phi = 0°, red) and H-plane cut (Theta = 90°, green). The E-plane pattern exhibits a well-defined broadside main lobe directed along theta = 0° (normal to the patch surface) with strong ground-plane suppression of back radiation, consistent with microstrip patch antenna theory. The H-plane pattern demonstrates near-omnidirectional behaviour in the azimuthal plane, characteristic of the TM_{11} resonant mode. The slight azimuthal asymmetry in the H-plane is attributed to the asymmetric stripline feed geometry. Maximum directivity of 7.09 dBi is achieved at broadside, confirming the antenna's suitability for broadside radiating applications



Maximum Delta S (convergence)

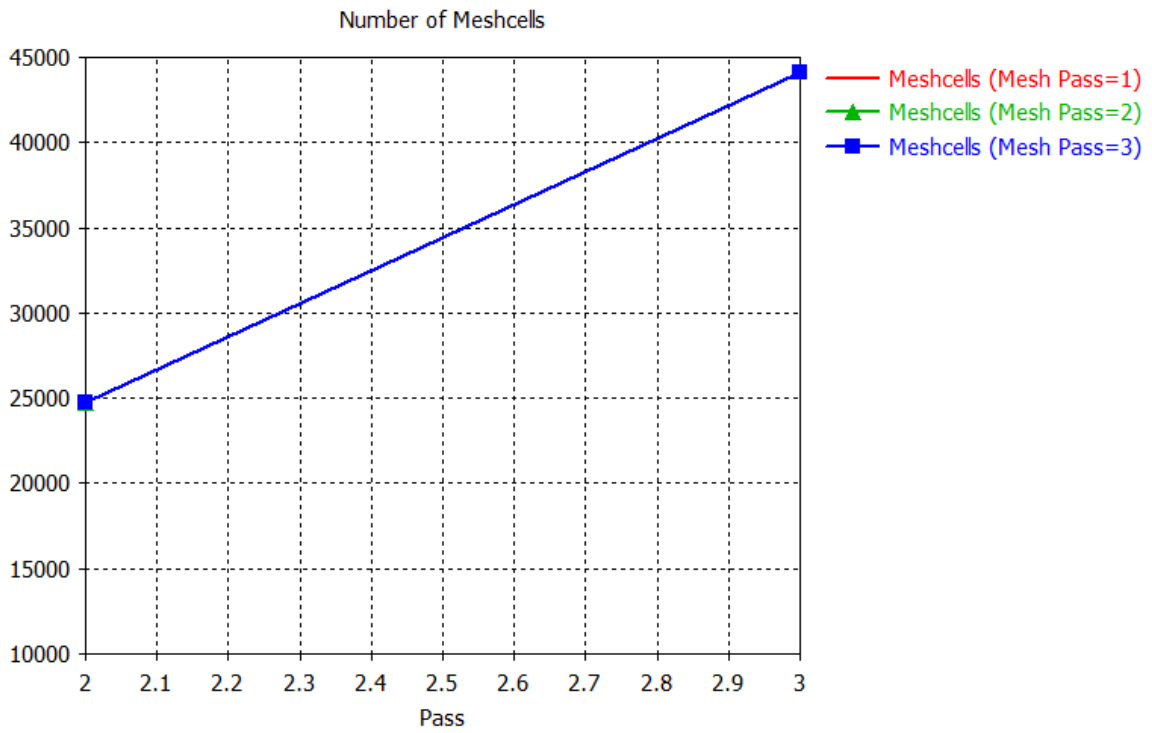
Maximum Delta S convergence history across adaptive mesh passes 2 and 3. Delta S represents the maximum change in S-parameter magnitude between successive mesh refinement passes and serves as the primary convergence criterion. The value decreases from 0.044 at Pass 2 to 0.005 at Pass 3 — an 89% reduction — confirming that the S-parameter solution has converged to a stable result. The final Delta S of 0.005 is well below the typical convergence threshold of 0.02, validating the accuracy of

the S11 and gain results obtained from the final mesh.



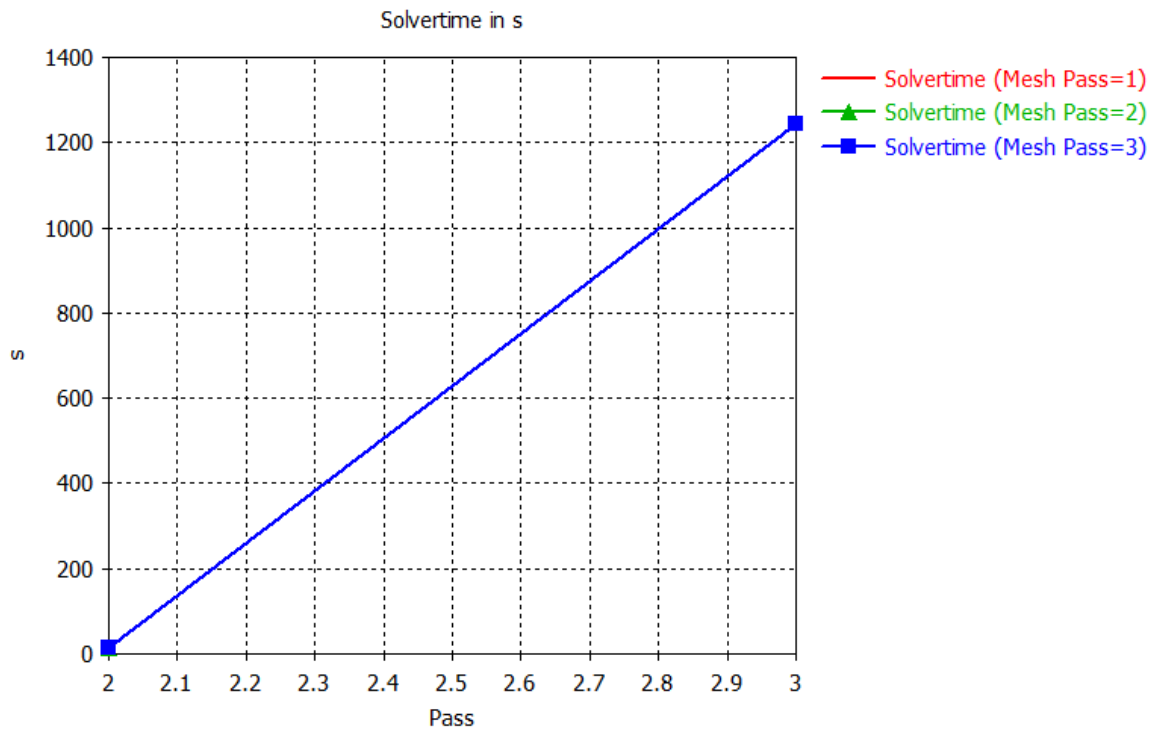
Number of Mesh Cells

Total number of mesh cells as a function of adaptive mesh pass. The mesh grows from approximately 25,000 cells at Pass 2 to 44,000 cells at Pass 3, representing a 76% increase in mesh density. This growth reflects the automatic refinement of critical regions including the circular patch perimeter, the stripline-to-patch junction, and the substrate-air interfaces where field gradients are steepest. The increasing mesh density combined with the converging Delta S and gain values in the preceding figures confirms that the simulation has achieved a good balance between accuracy and computational cost.



Solver Time

Solver computation time as a function of adaptive mesh pass. Solve time increases from approximately 25 seconds at Pass 2 to 1,240 seconds (approximately 21 minutes) at Pass 3, corresponding to a 50-fold increase in computational cost driven by the 76% growth in mesh cell count between passes. This nonlinear scaling is consistent with the time-domain finite integration technique (FIT) solver, where computation time scales superlinearly with the number of mesh cells. The final solve time of approximately 21 minutes represents an acceptable computational cost for the convergence quality achieved — Delta S = 0.005 and gain variation < 0.1 dBi between passes


Solvertime in s
1400
1200
1000
800
600
400
200
0
s
2
2.1
2.2
2.3
2.4
2.5
2.6
2.7
2.8
2.9
3
Pass
Solvertime (Mesh Pass=1)
Solvertime (Mesh Pass=2)
Solvertime (Mesh Pass=3)